Notes

Assignments

4.50 & 4.51 attached

Tutorial problems

7.41, 7.44, 7.47, 7.49

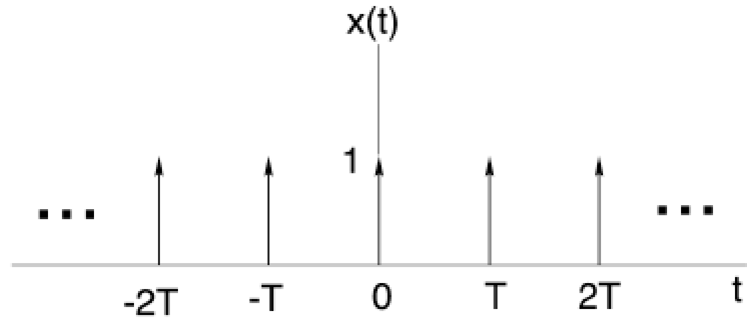
Outline

- Sampling is a general procedure to generate DT signals from CT signals, where information of the original signals can be kept
- Core sampling theory:
  - Impulse train, zero-order hold, 1st-order hold, etc.
  - Analysis in frequency domain
  - Nyquist rate
- Undersampling: Aliasing
- Application: process continuous-time signals discretely
- More sampling techniques: decimation, downsampling and upsampling

Review

Example 4.8

$$x(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad \text{— sampling function}$$



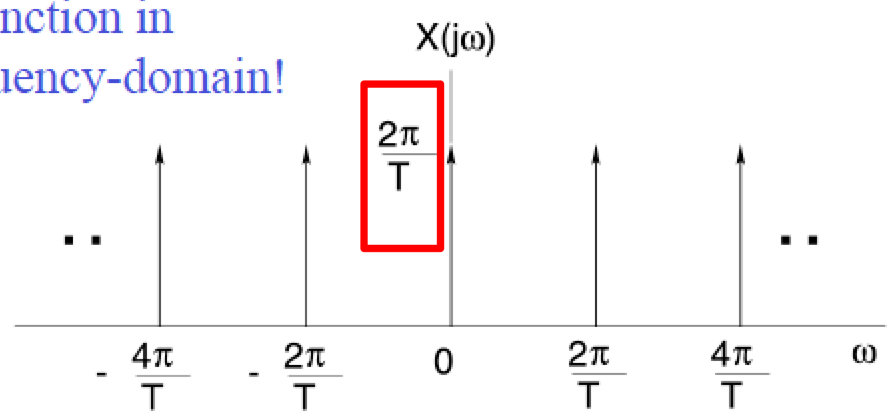
$$x(t) \overset{FS}{\longleftrightarrow} a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-jk\omega_o t} dt = \frac{1}{T}$$

$$\Downarrow \; x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T} e^{jk\omega_o t}$$

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} \boxed{\frac{2\pi}{T}} \delta(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_o})$$

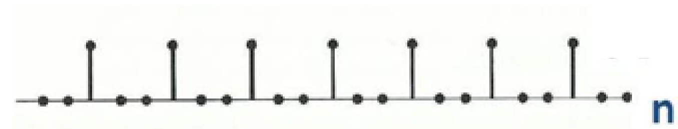
$\omega_s = 2\pi / T$: sampling frequency

Same function in
the frequency-domain!



Example: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n-kN]$?
- First of all, we calculate the Fourier series:



$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\frac{2\pi}{N}n} \\ &= \frac{1}{N} \sum_{n=<N>} \sum_{k=-\infty}^{+\infty} \delta[n-kN] e^{-jk\frac{2\pi}{N}n} \\ &= \frac{1}{N} \sum_{n=<N>} \delta[n] e^{-jk\frac{2\pi}{N}n} = \frac{1}{N} \end{aligned}$$

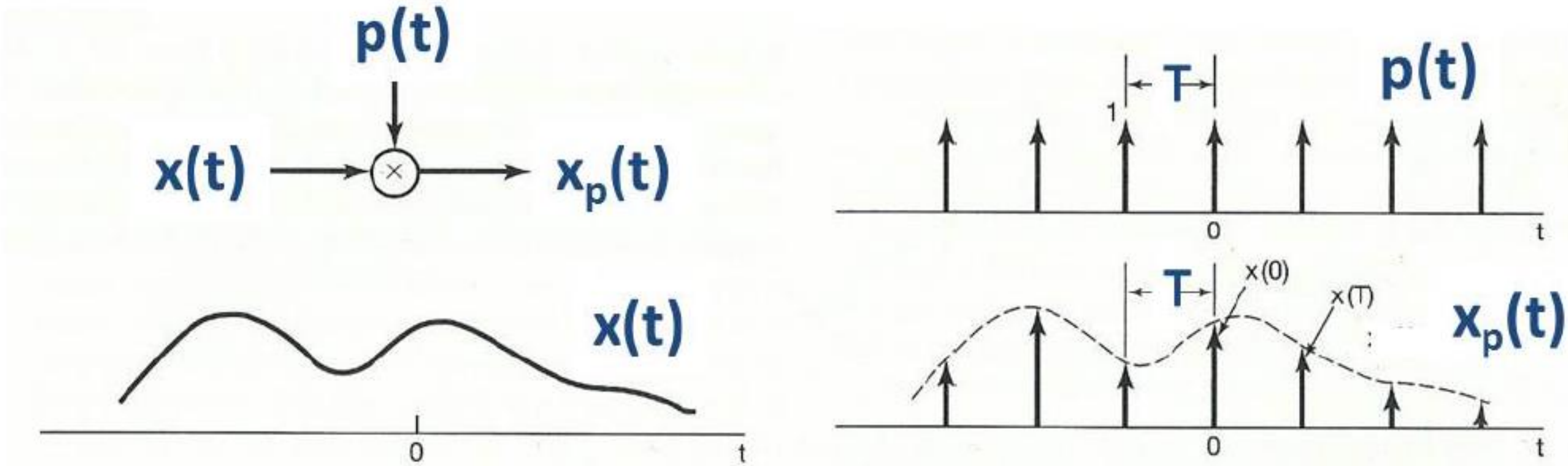
- Then, we have

$\omega_s = 2\pi / N$: sampling frequency

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k(2\pi/N) - 2\pi l) = \boxed{\frac{2\pi}{N}} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

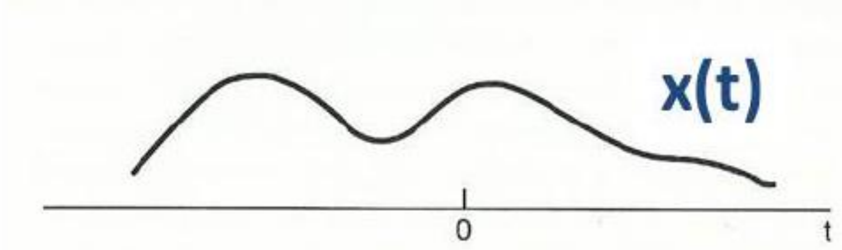
Impulse-train sampling

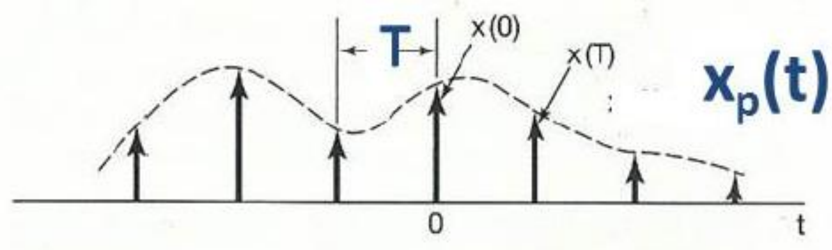
- Mathematically, sampling can be represented by multiplication

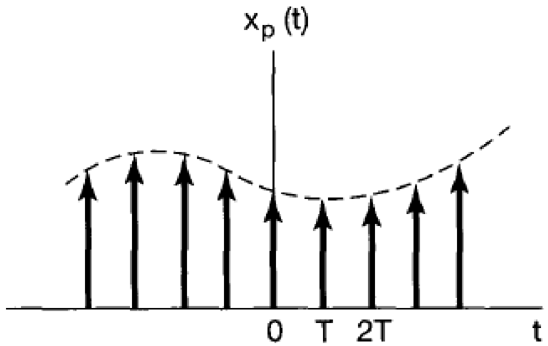


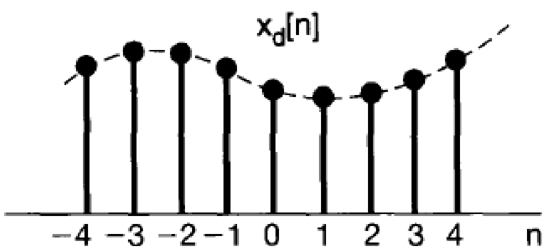
- Sampling function: $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$
- Sampling period: T
- Sampling:

$$x_p(t) = x(t) \times p(t) = x(t) \times \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$$







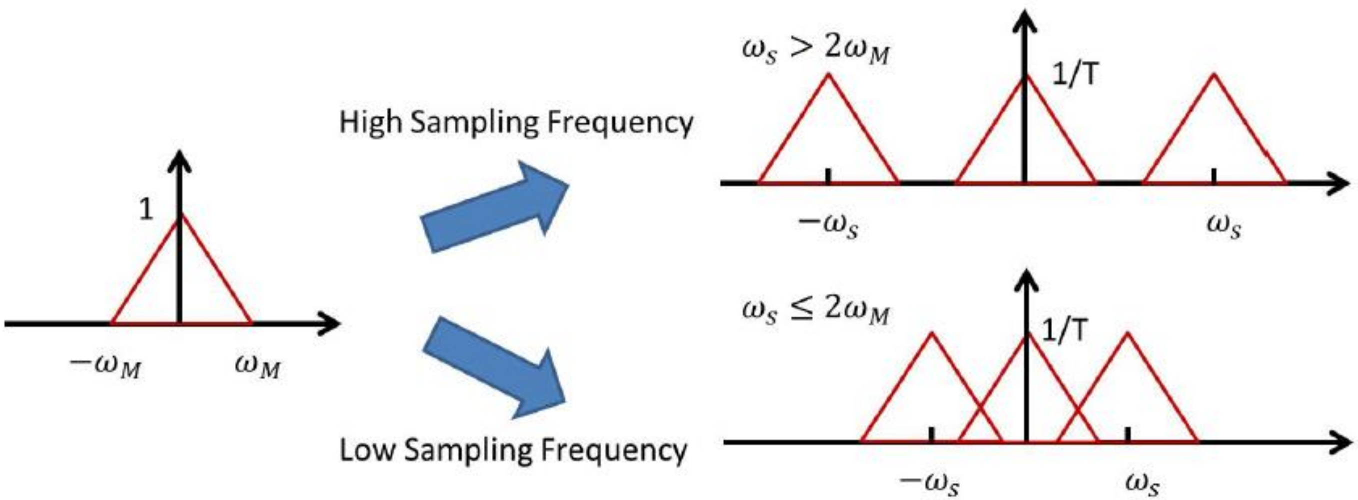


$$\boxed{x_d[n] = x(nT)}$$

x(t): CT → x_p(t): CT → x_d[n]: DT

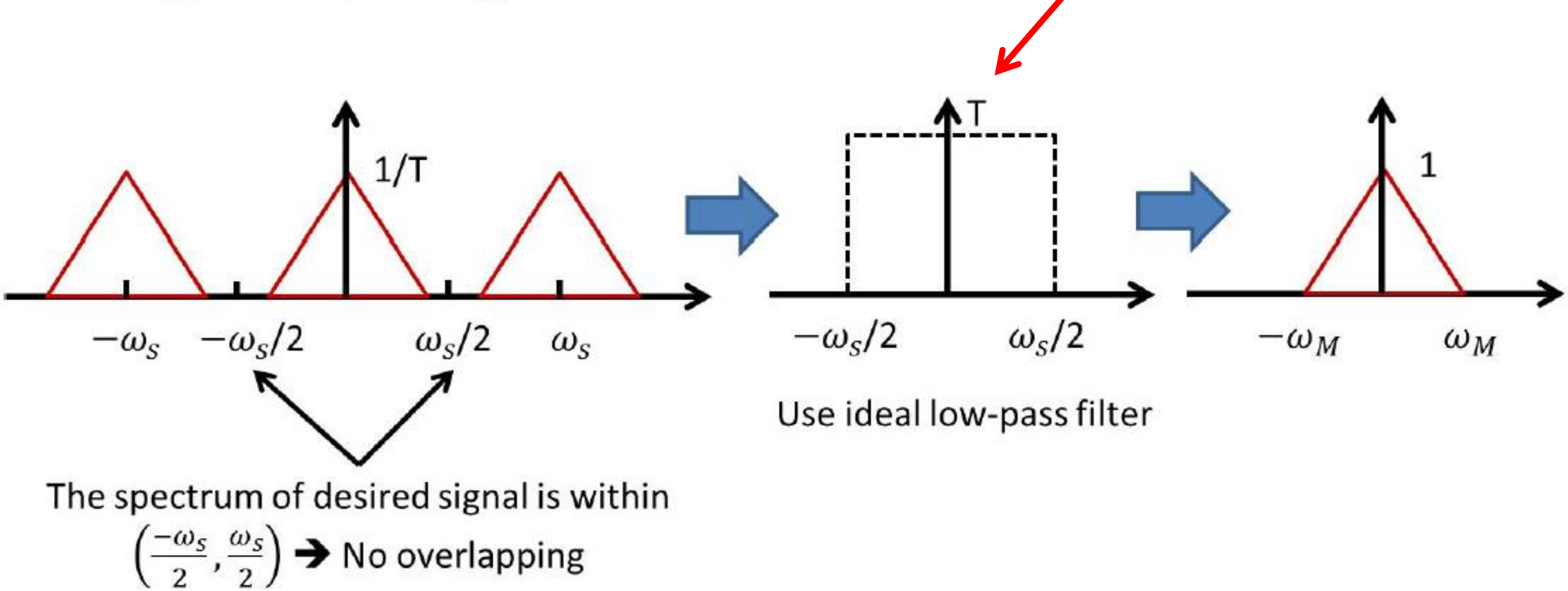
Frequency analysis

- Sampling: the Fourier transform of input signal is repeated with period ω_s



Reconstruction

- Scenario of $\omega_s > 2\omega_M$



Sampling theorem

Sampling Theorem

Let $x(t)$ be a band-limited signal with

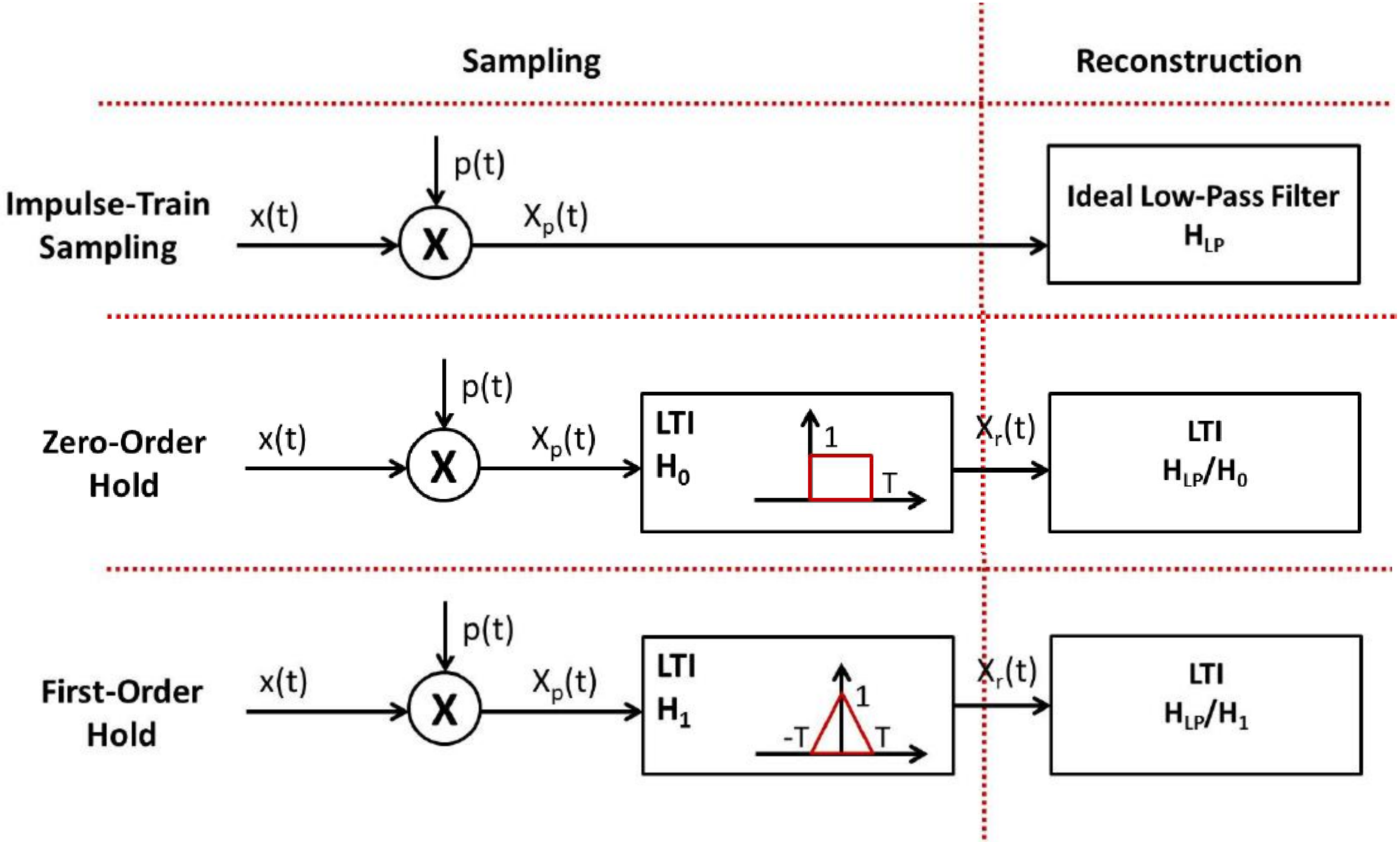
$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M.$$

Then, $x(t)$ is uniquely determined by its samples $x(nT)$ or $x_p(t)$ if

$$\omega_s = \frac{2\pi}{T} > 2\omega_M,$$

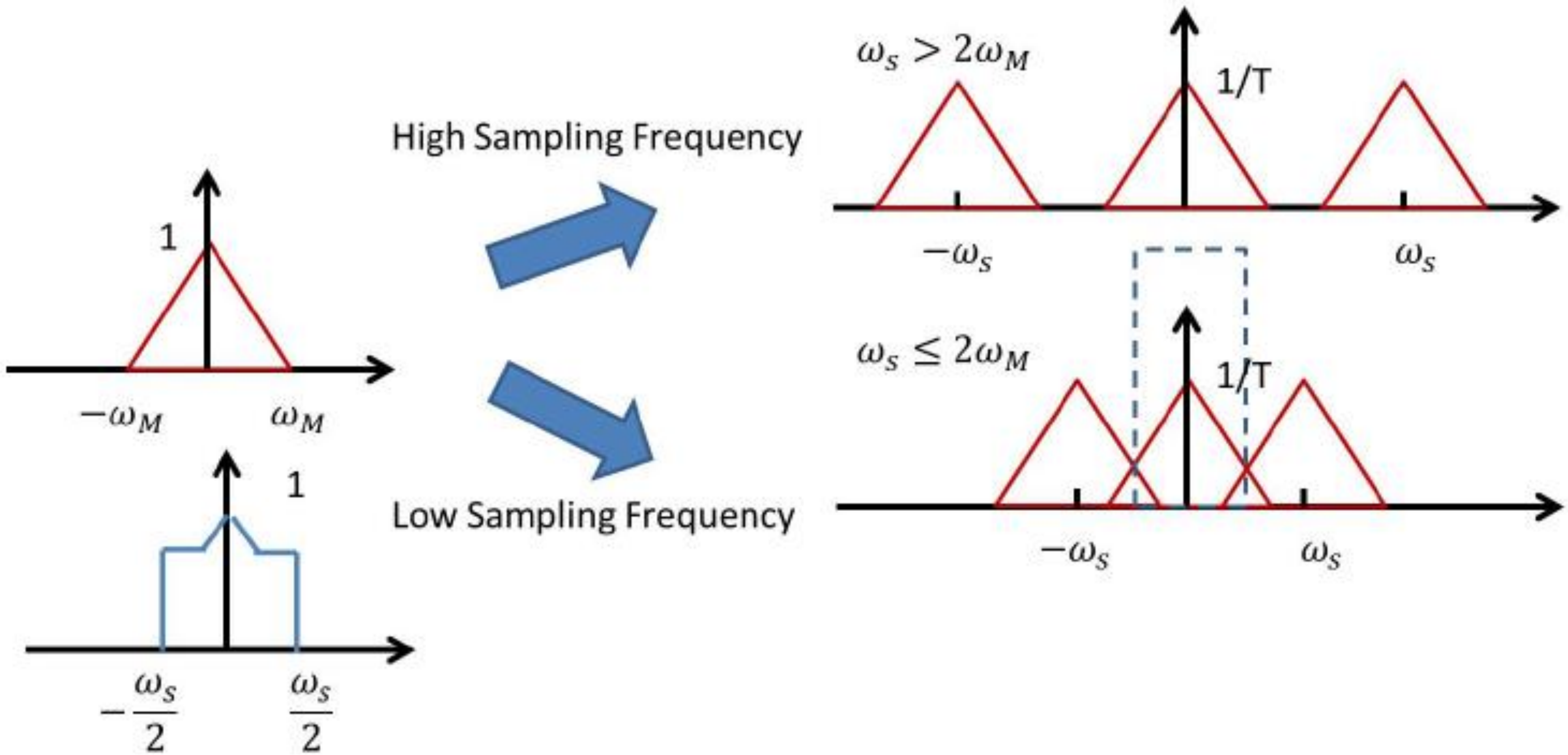
where $2\omega_M$ is referred to as the Nyquist rate.

Summary: Sampling approaches



Undersampling & Aliasing

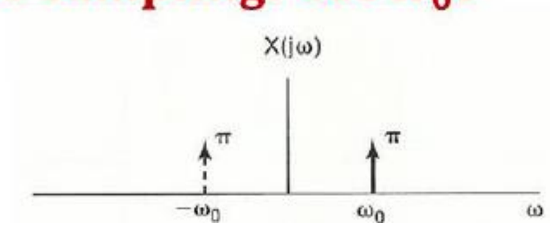
- Undersampling: insufficient sampling frequency $\omega_s < 2\omega_M$
- Perfect reconstruction is impossible with undersampling.
- Aliasing: distortion due to undersampling



Aliasing: Example

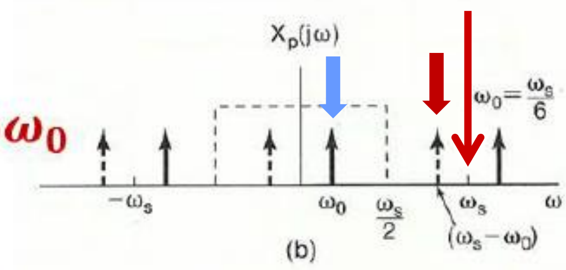
Signal before sampling: $\cos \omega_0 t$

Sampling rate: ω_s

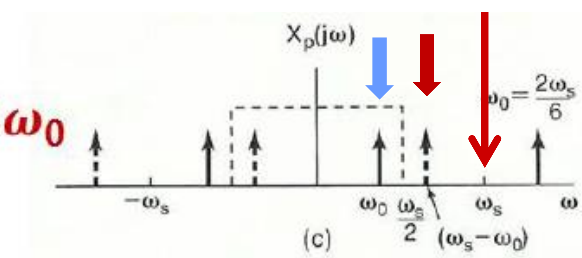


Lowpass Filter: $\dfrac{-\omega_s}{2} \sim \dfrac{\omega_s}{2}$

$\omega_s = 6\omega_0$



$\omega_s = 3\omega_0$

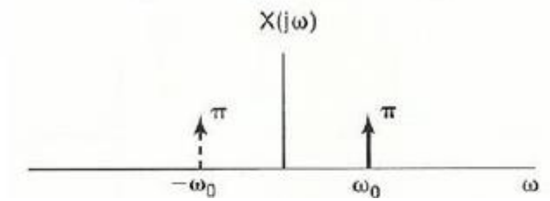


$\cos \omega_0 t$

Aliasing: Example

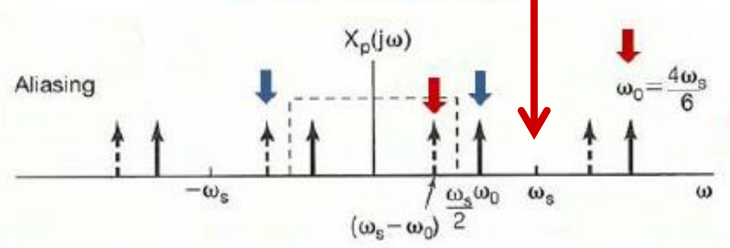
Signal before sampling: $\cos \omega_0 t$

Sampling rate: ω_s



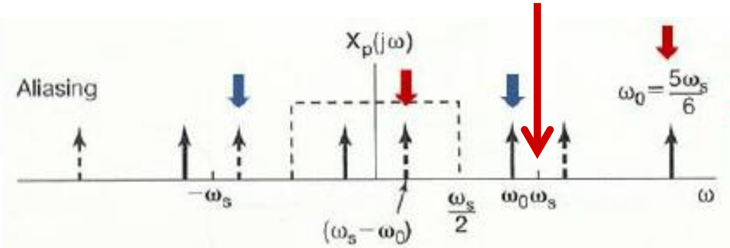
Lowpass Filter: $\frac{-\omega_s}{2} \sim \frac{\omega_s}{2}$

Undersampling



(d)

$$\omega_s = \frac{3}{2}\omega_0$$



(e)

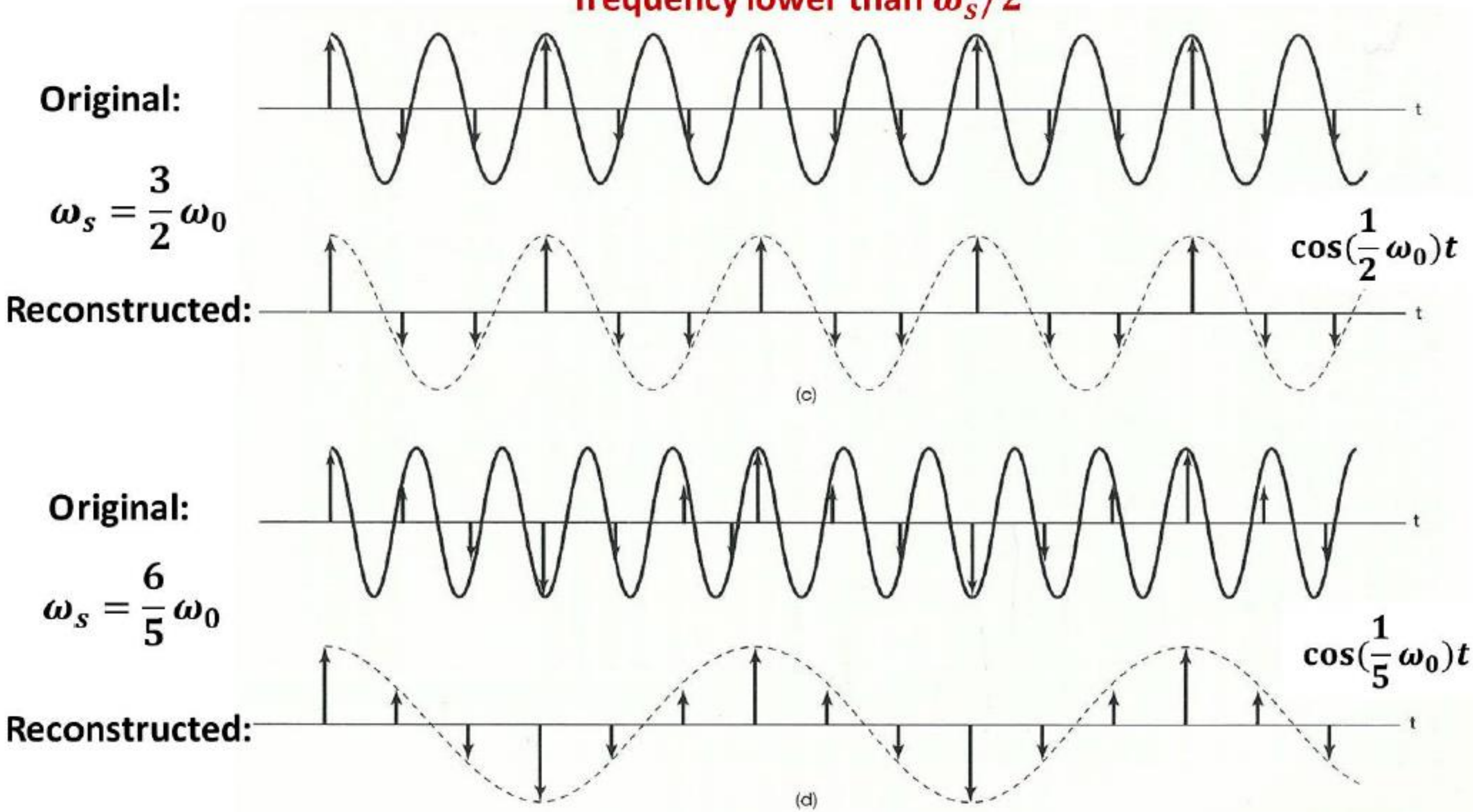
$$\omega_s = \frac{6}{5}\omega_0$$

Aliasing: $\cos(\omega_s - \omega_0)t$

Under fixed w_s, when w_0 is increasing, what is the frequency of recovered wave?

Aliasing: Example

Low-pass filtering: Interpret the samples by cosine function with frequency lower than $\omega_s/2$

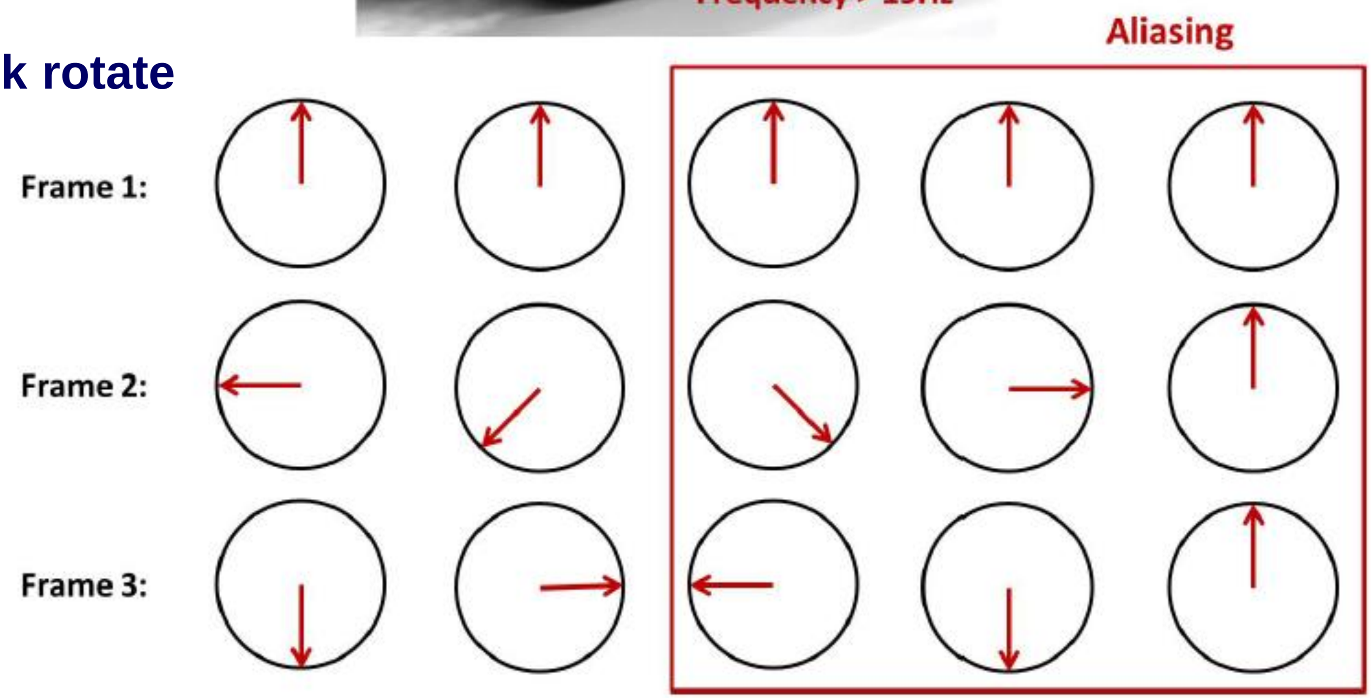


Aliasing: $\cos(\omega_s - \omega_0)t$

Aliasing in Movies

- Wheel's rotation in movies

Anti-clock rotate

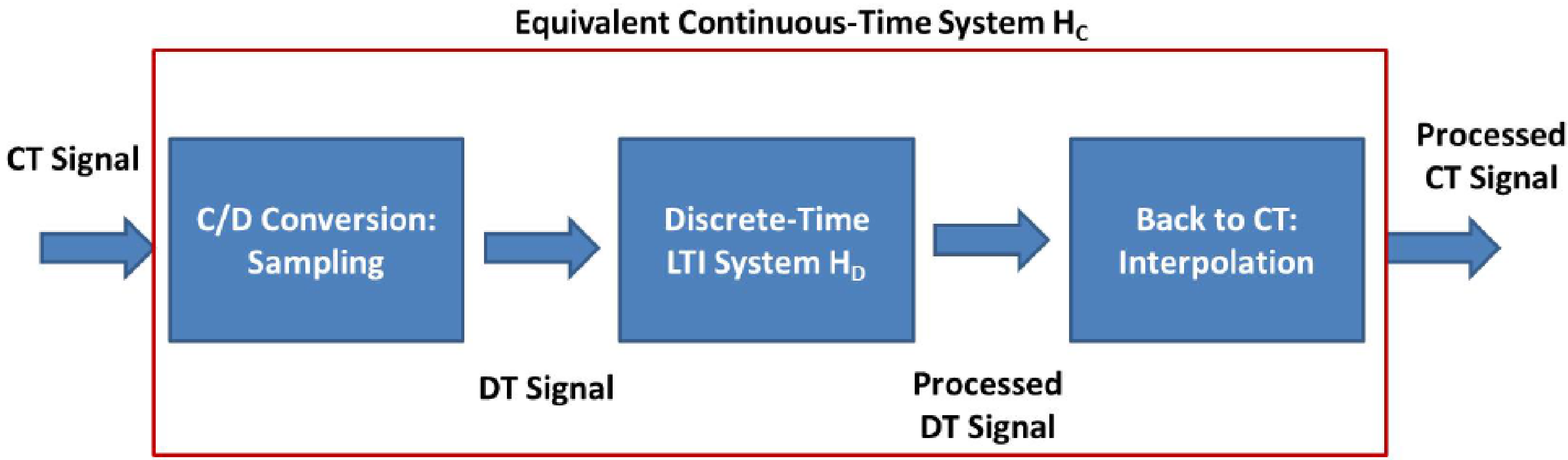


90^0 / frame 135^0 225^0 270^0 360^0

Process Continuous-Time Signals Discretely

- People would like to process continuous-time signal in discrete-time (digital) domain

Block Diagram



- It is much easier to design DT system.
- What's the relation between H_C and H_D?

or

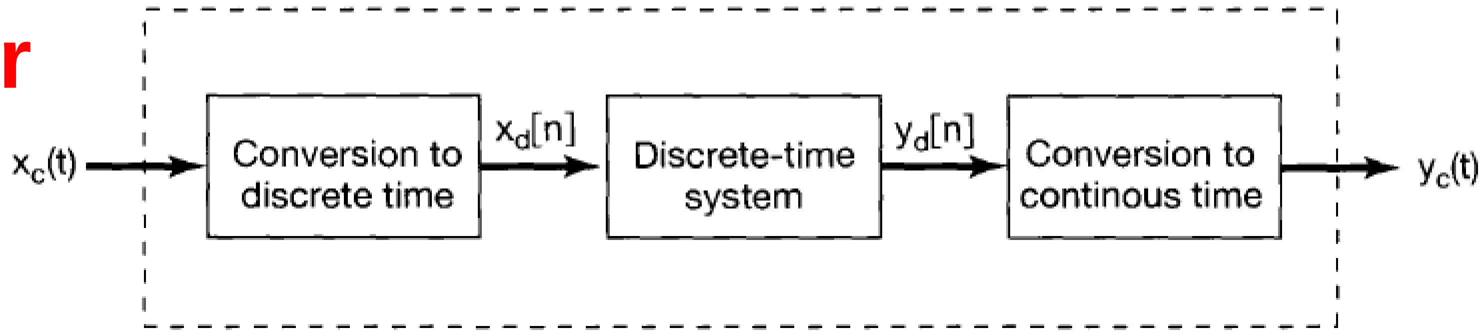


Figure 7.19 Discrete-time processing of continuous-time signals.

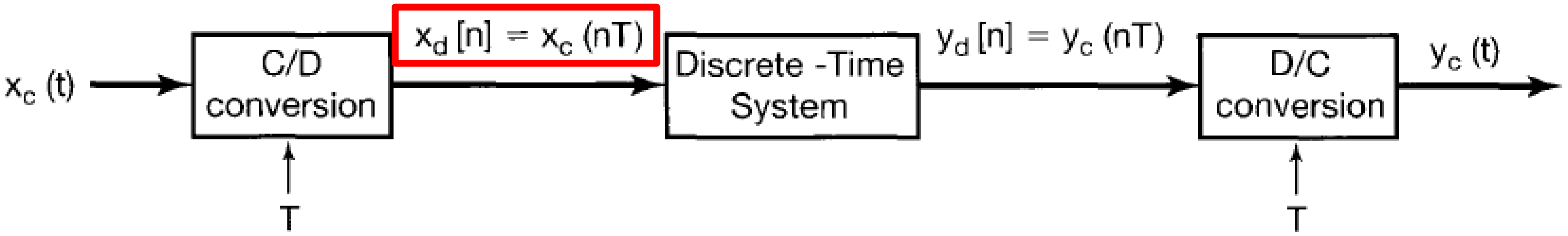
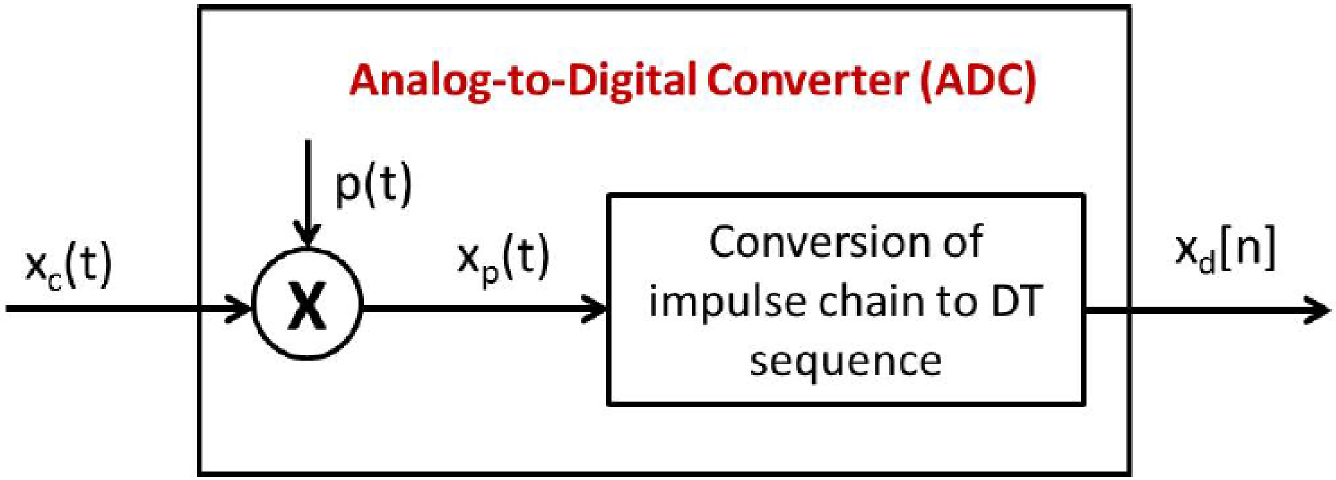


Figure 7.20 Notation for continuous-to-discrete-time conversion and discrete-to-continuous-time conversion. T represents the sampling period.

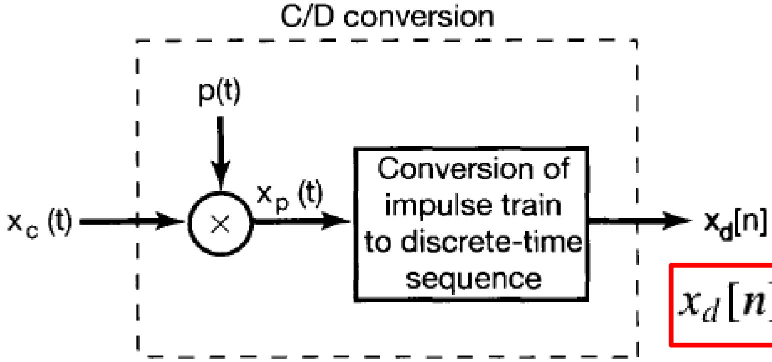
Discretization: C/D Conversion



- Mathematical Interpretation (Fourier Transform)

$$x_c(t) \quad \longleftrightarrow \quad X_c(j\omega)$$

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT) \quad \longleftrightarrow \quad X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$



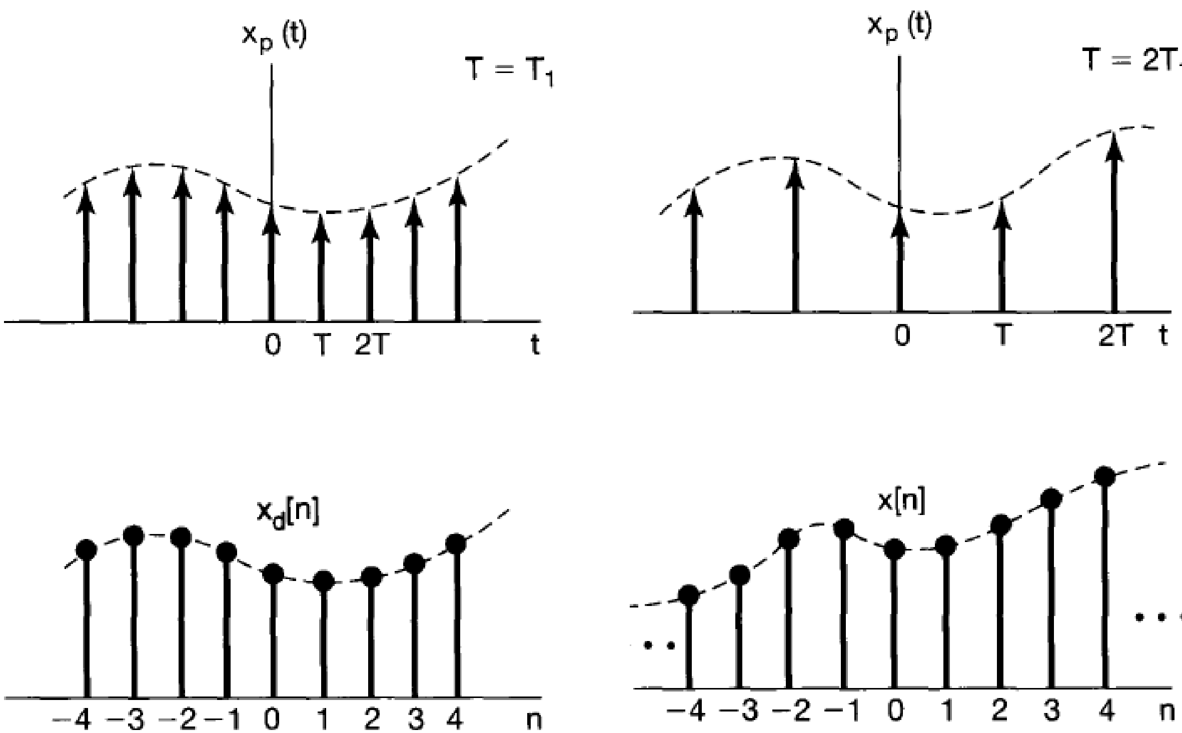
$$\boxed{x_d[n] = x_c(nT)}$$

(a)

$$X_p(j\omega) = \sum_{t=-\infty}^{\infty} x_p(t)e^{-j\omega t}$$

$$= \sum_{n=-\infty}^{\infty} [x_c(nT)\delta(t-nT)]e^{-j\omega t}$$

$$= \sum_{n=-\infty}^{\infty} x_c(nT)e^{-j\omega nT}$$



$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n}$$

$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$

$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

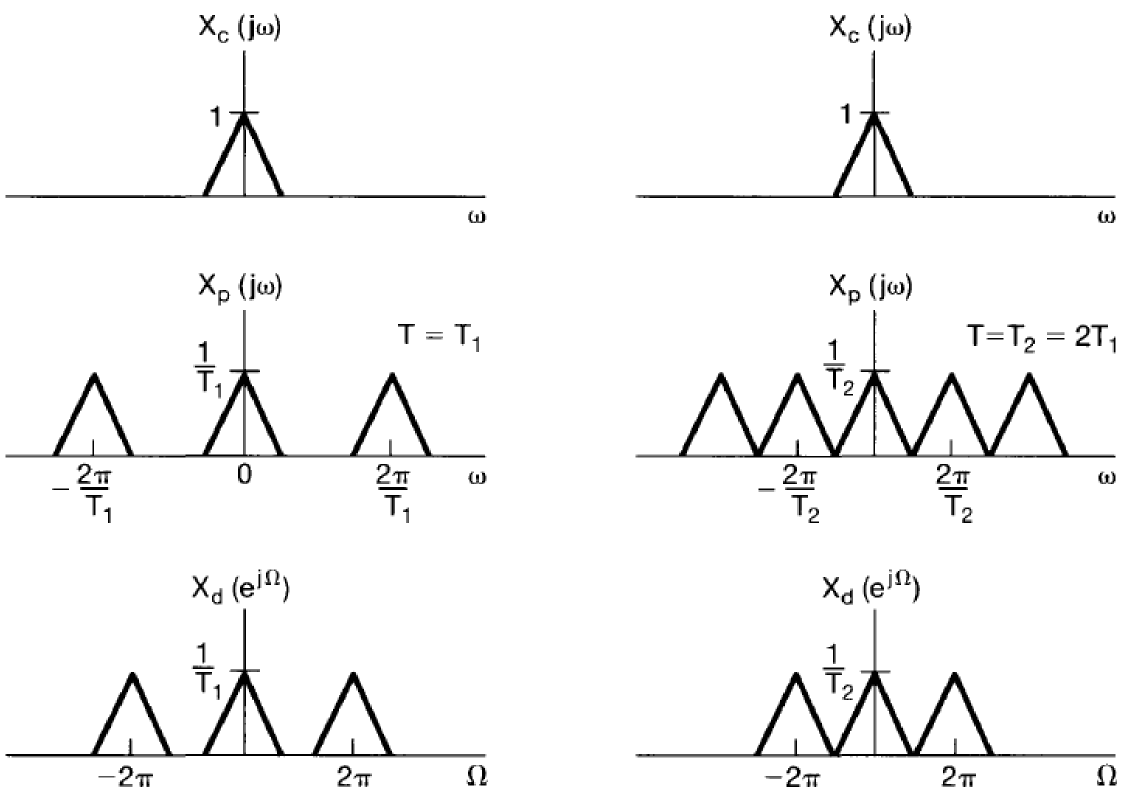


Figure 7.22 Relationship between $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ for two different sampling rates.

1. **pulse-train sampling**

2. scaling (expansion)

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$

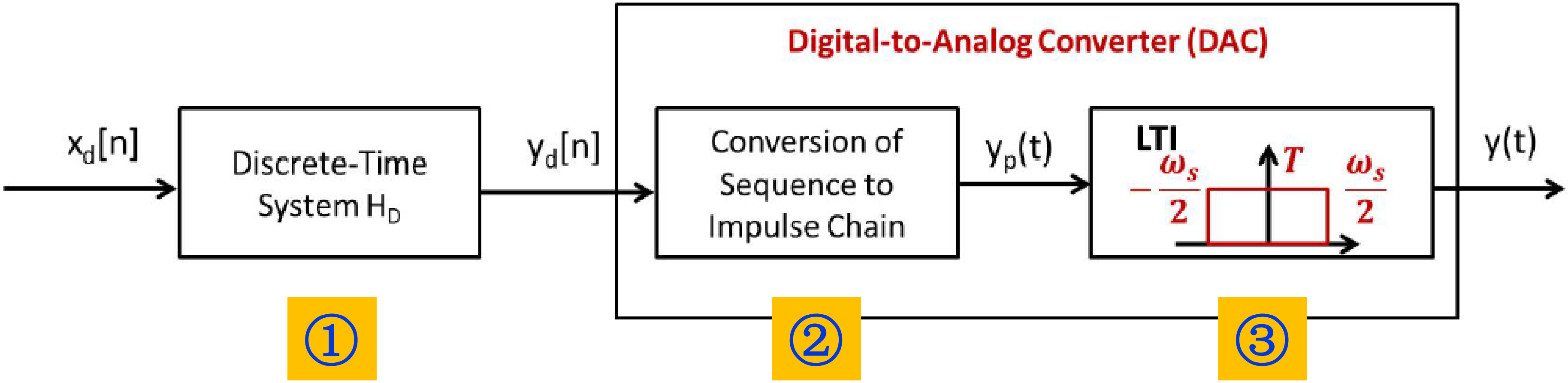
$$X_d(e^{j\omega}) = X_p(j\omega / T)$$

$$\Downarrow$$

$$X_d(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega - 2\pi k}{T}))$$

DT Processing and Conversion



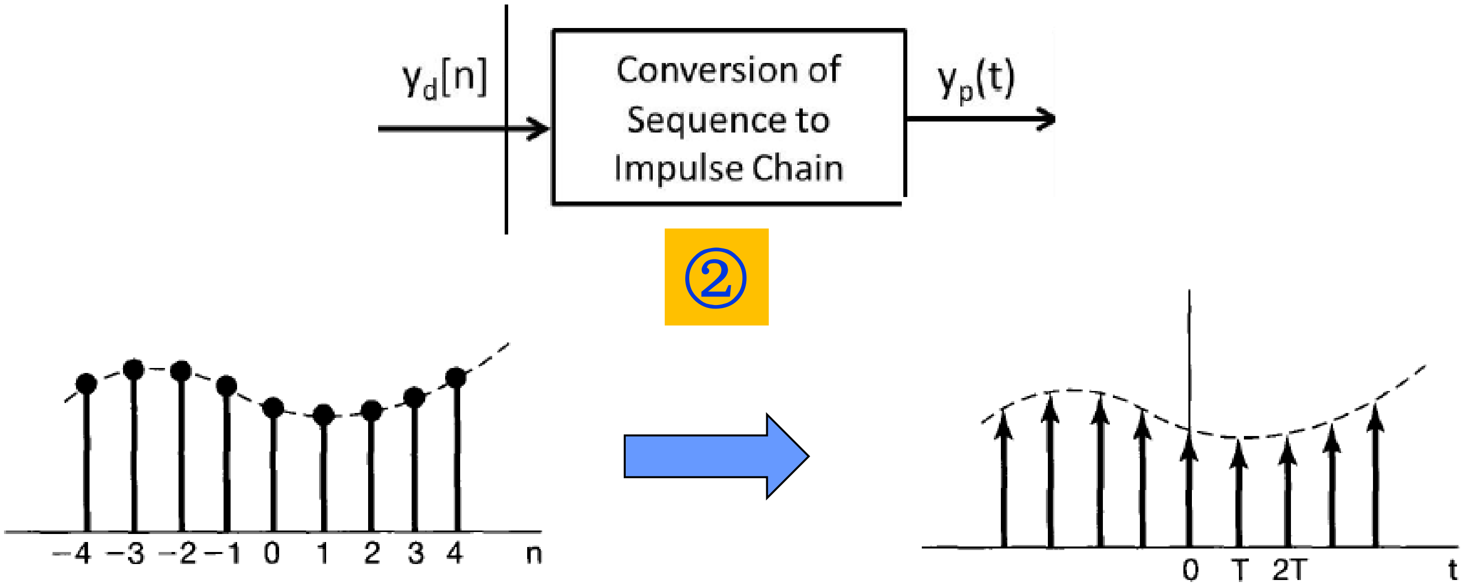
- Mathematical Interpretation (Fourier Transform)

① $$y_d[n] = x_d[n] * h_D[n] \quad \longleftrightarrow \quad Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$$

② $$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$

③ $$y(t) = y_p(t) * h_{LP}(t) \quad \longleftrightarrow \quad Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$$

$y_d[n] = y_c(nT)$

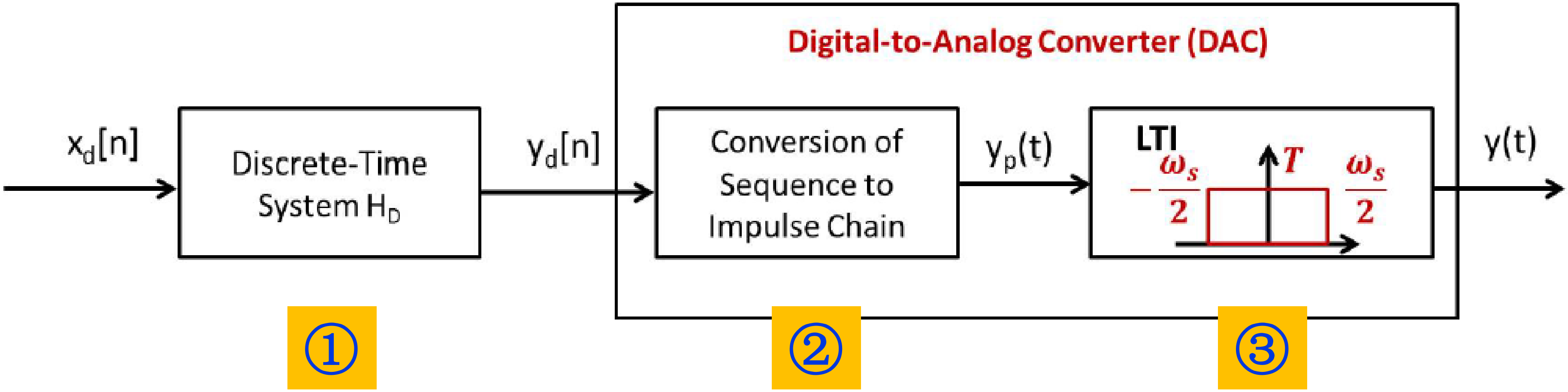


1. **n → t**
2. **time-scaling**

$$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$

Prove this !

DT Processing and Conversion



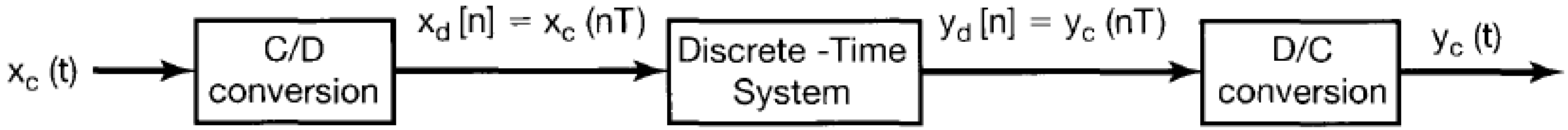
- Mathematical Interpretation (Fourier Transform)

① $$y_d[n] = x_d[n] * h_D[n] \quad \longleftrightarrow \quad Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$$

② $$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$

③ $$y(t) = y_p(t) * h_{LP}(t) \quad \longleftrightarrow \quad Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$$

$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$



$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$X_d(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

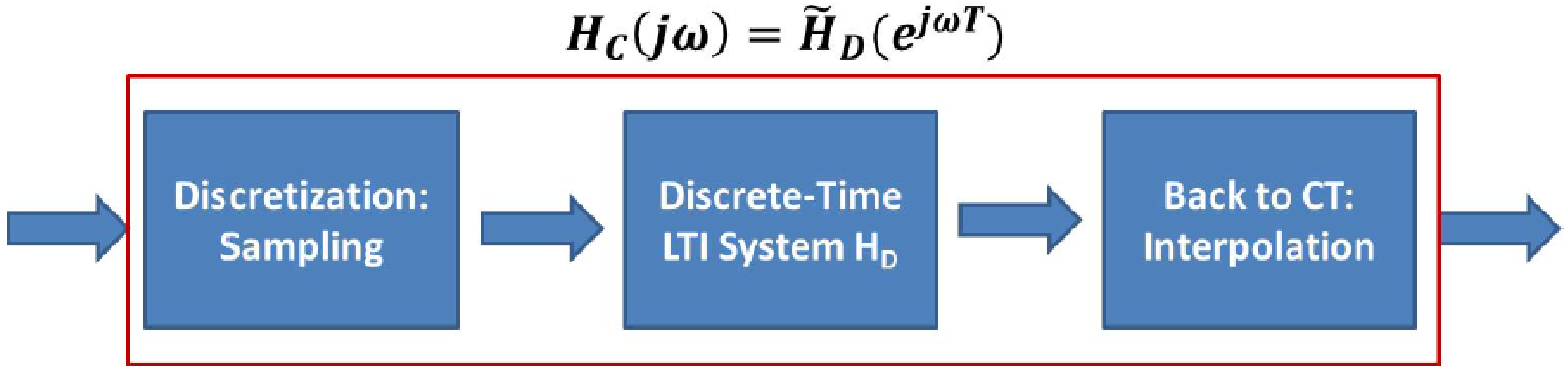
$$\begin{aligned} Y(j\omega) &= \left[\frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))\right] H_D(e^{j\omega T})H_{LP}(j\omega) \\ &= X_c(j\omega)H_D(e^{j\omega T}) \\ &= X_c(j\omega)\widetilde{H}_D(e^{j\omega T}) \qquad (1) \end{aligned}$$

where

$$\widetilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & otherwise \end{cases} \qquad (2)$$

- It is equivalent to a continuous-time LTI system $H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})$
- $H_D(e^{j\omega T})$ is a periodic extension of $\widetilde{H}_D(e^{j\omega T})$ with period $\omega_s = 2\pi/T$

System Design

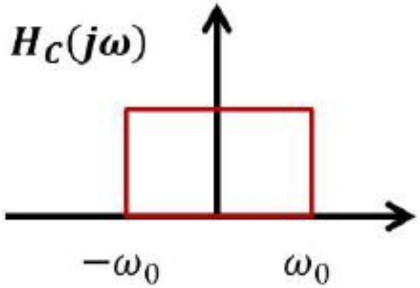


- How can we design a CT LTI system with frequency response H_C via DT LTI system?
- Step 1: Sampling frequency ω_s or $2\pi/T$ should be larger than Nyquist rate
- Step 2: $\widetilde{H}_D(e^{j\omega T}) = H_C(j\omega)$
- Step 3: Frequency response of DT LTI system
  $H_D(e^{j\omega T}) = \sum_{k=-\infty}^{\infty} \widetilde{H}_D(e^{j(\omega - k\omega_s)T})$ or
  $H_D(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \widetilde{H}_D(e^{j(\omega - k\omega_s T)}) = \sum_{k=-\infty}^{\infty} H_C(j\frac{\omega - 2k\pi}{T})$

System Design Example

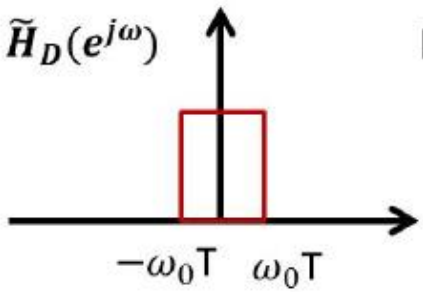
- How to implement an ideal CT lowpass filter?

Objective of Design:

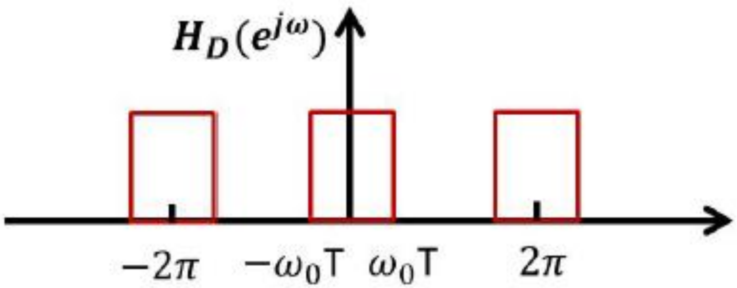


$$H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})$$

Scale by T

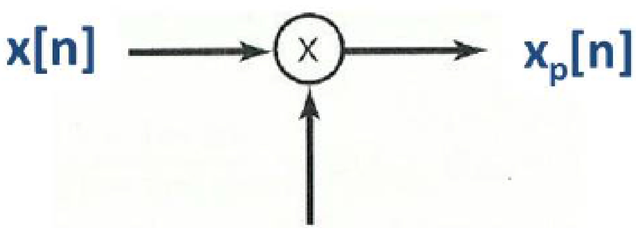


Repetition



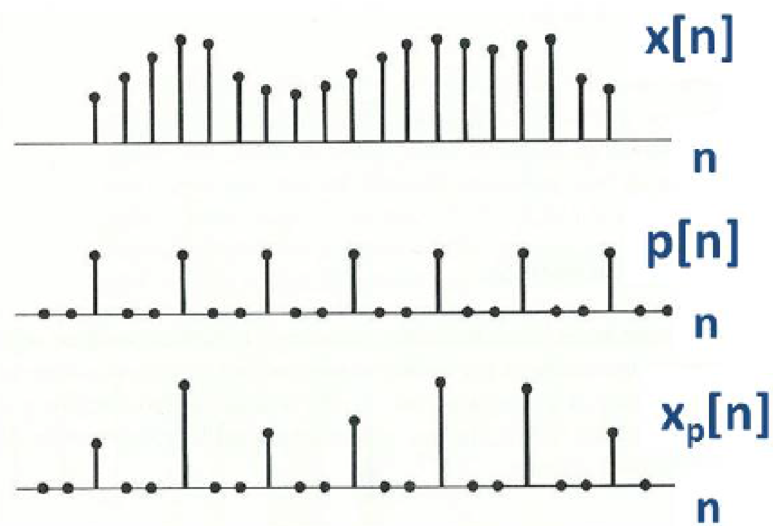
Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$\mathbf{p[n]} = \sum_{\mathbf{k}=-\infty}^{+\infty} \boldsymbol{\delta}[\mathbf{n}-\mathbf{kN}]$$

$$\mathbf{x_p[n]} = \sum_{\mathbf{k}=-\infty}^{+\infty} x[kN]\boldsymbol{\delta}[\mathbf{n}-\mathbf{kN}]$$



N

Frequency Analysis

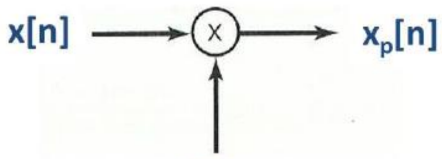
$$P(e^{j\omega}) = \frac{2\pi}{N}\sum_{k=-\infty}^{\infty}\delta(\omega - k\omega_s) \quad where \boxed{\omega_s = 2\pi/N}$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} P(e^{j\theta})X(e^{j(\omega-\theta)})d\theta = \boxed{\frac{1}{N}}\sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$

periodic convolution

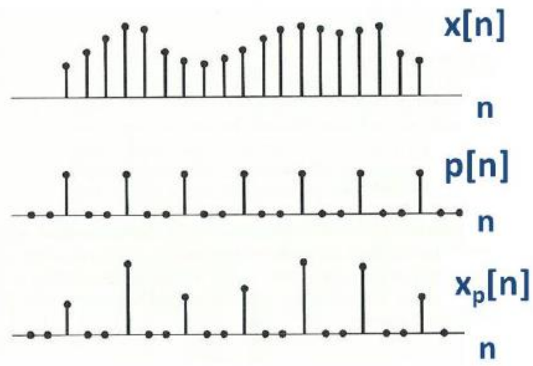
$$= \frac{1}{N}[X(e^{j\omega}) + X(e^{j(\omega-\omega_s)}) + \ldots + X(e^{j(\omega-(N-1)\omega_s)})]$$

What is the relation between $X(e^{j\omega})$ and $X_p(e^{j\omega})$?



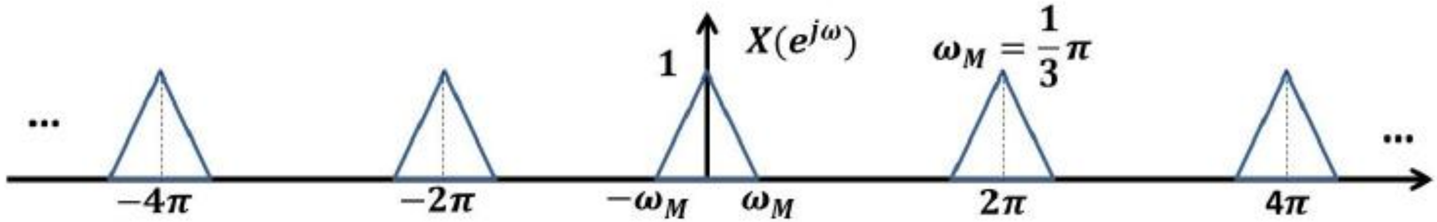
$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n-kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN]\delta[n-kN]$$



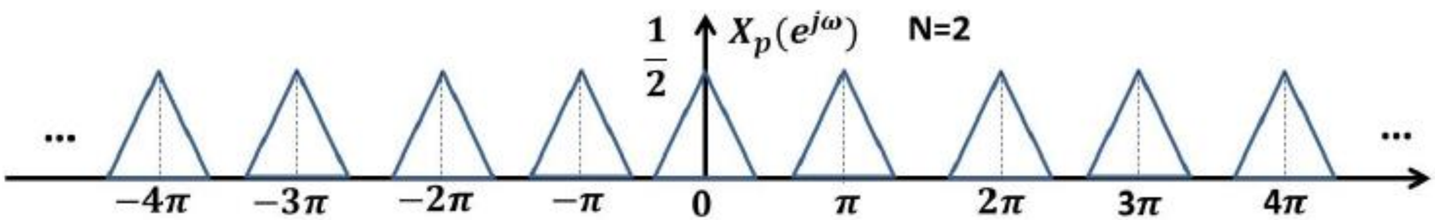
N

$\boxed{\omega_s = 2\pi}$



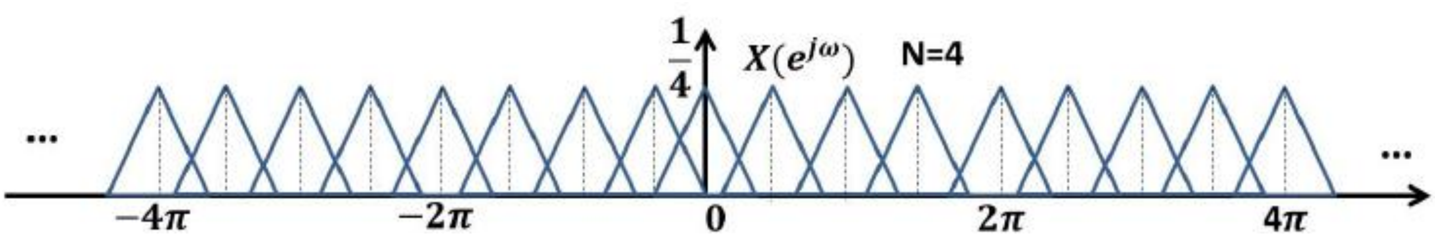
N=2

$\boxed{\omega_s = \pi}$



N=4

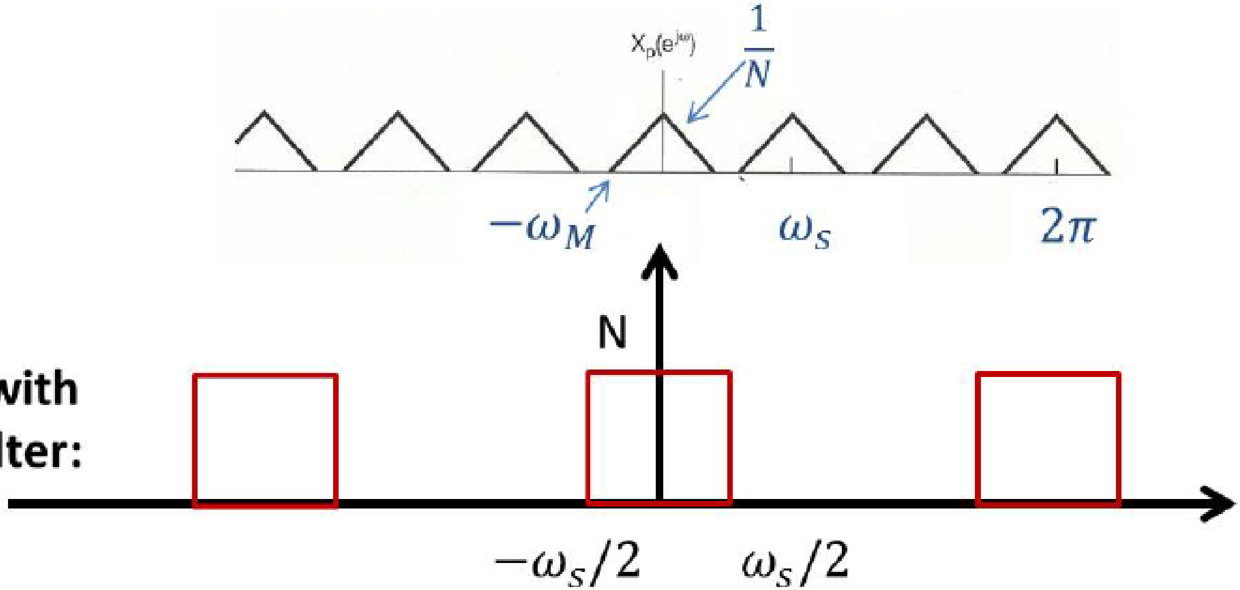
$\boxed{\omega_s = \frac{\pi}{2}}$



Reconstruction

- Perfect reconstruction is applicable when $\omega_s > 2\omega_M \leftrightarrow N < \frac{\pi}{\omega_M}$

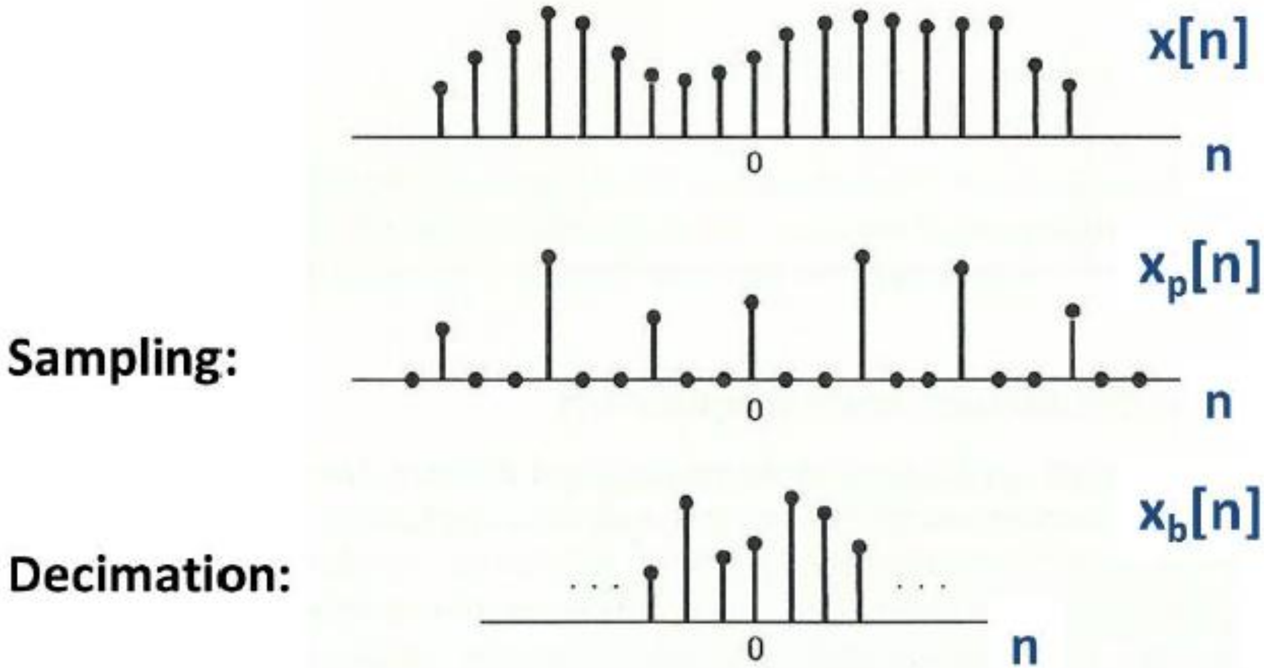
Reconstruction with ideal low-pass filter:



- Aliasing occurs when $\omega_s < 2\omega_M$

Decimation

- After sampling, there will be a great amount of redundancy
- Decimation: discrete-time sampling + remove zeros



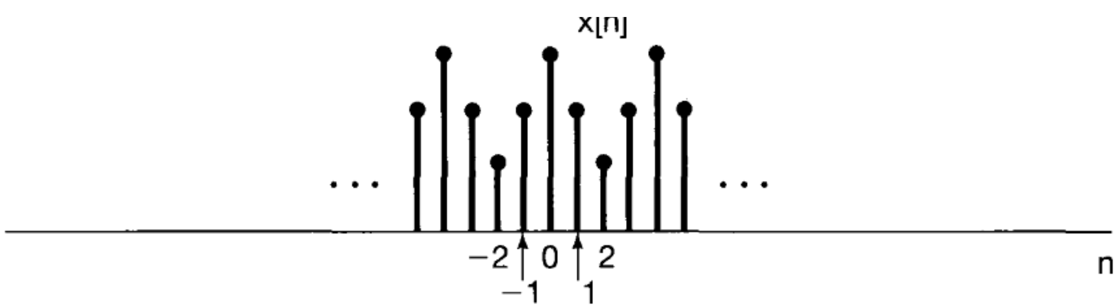
$$x_b[n] = x_p[nN] = x[nN]$$

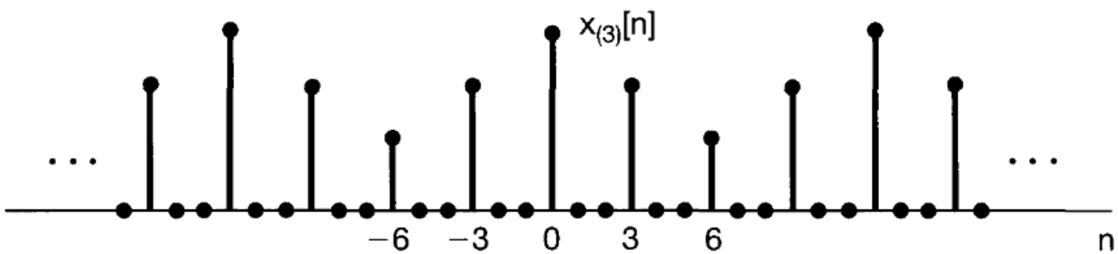
What is the relation between $X(e^{j\omega})$ and $X_b(e^{j\omega})$?

Review

- Time Expansion
  - Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of k} \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$





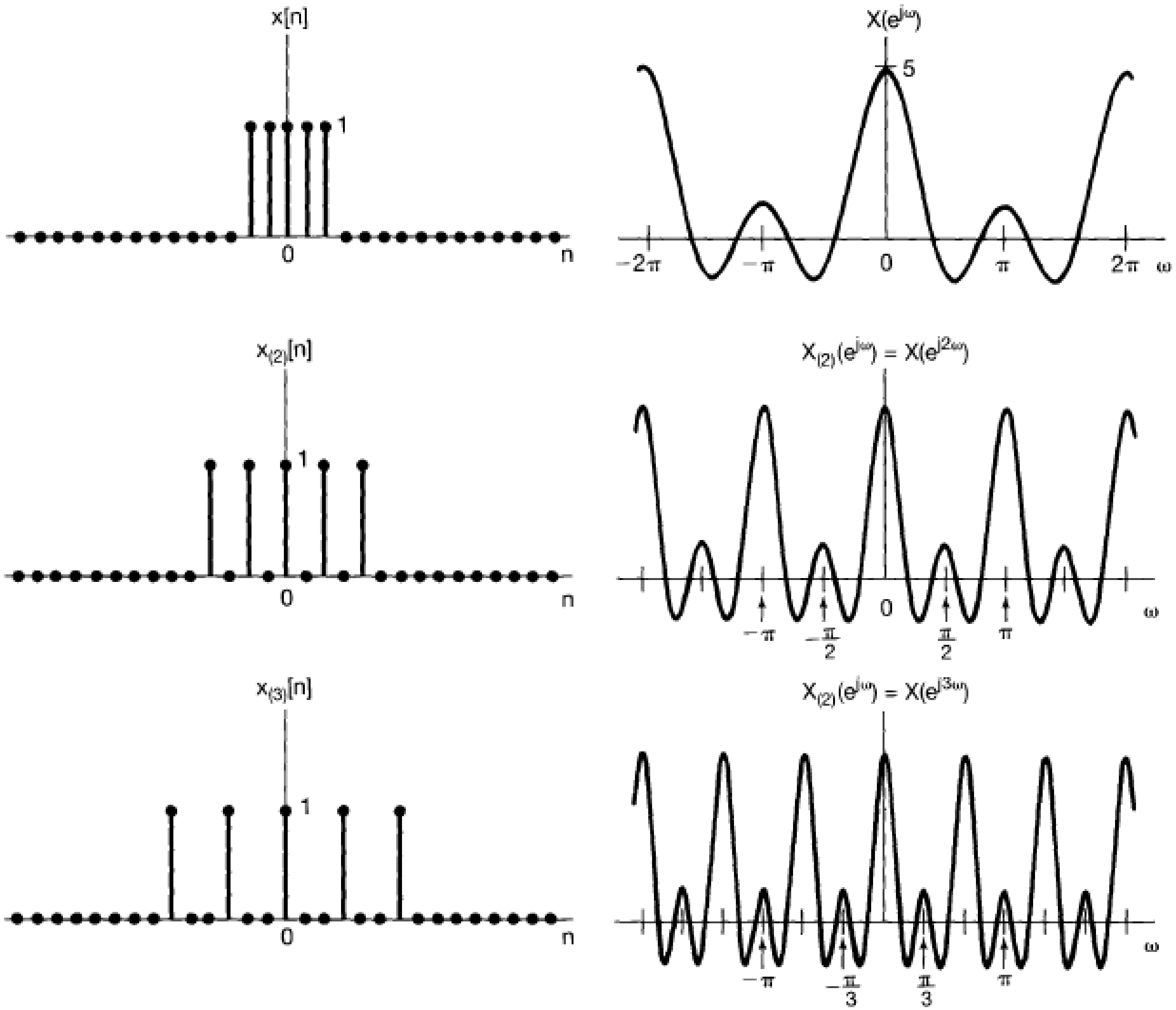


Figure 5.14 Inverse relationship between the time and frequency domains: As k increases, $x_{(k)}[n]$ spreads out while its transform is compressed.

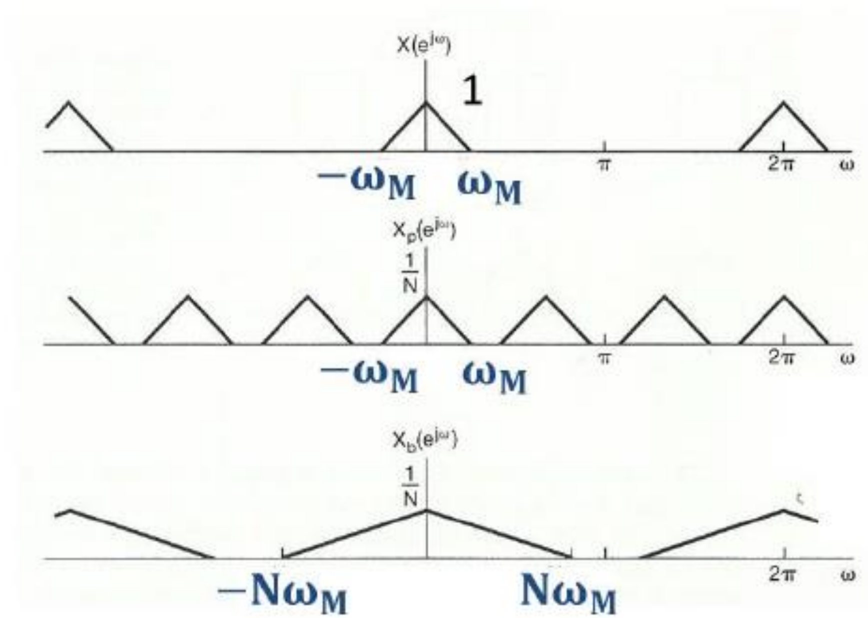
Frequency Analysis

$$X_b(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_b[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N})$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j\omega/N-k\omega_s})$$

$$\boxed{X_p(e^{j\omega}) = \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})}$$

$$\boxed{\omega_s = 2\pi/N}$$



$\boxed{\omega_s = 2\pi}$

Sampling: $\boxed{\omega_s = \dfrac{2\pi}{N}}$

Decimation: $\boxed{\omega_s = 2\pi}$

Condition for Perfect Reconstruction: $N\omega_M < \pi$

Summary

- Undersampling
  - Aliasing
- How to process CT signal with DT system?

$$H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})$$

- Sampling on DT signal